

Mutual Fund Analytics Platform

A Full-Stack Data Analytics Capstone Project

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Repository	github.com/Shahinshafi1717/mutual-fund-platform

Python 3.14

SQLite

Power BI

Pandas

NumPy

Plotly

₹62.7L Cr

Total AUM

107,779

DB Rows

40

Funds

11

DB Tables

7 Days

Built In

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1. Executive Summary

This report presents the complete findings of a 7-day capstone project building a Mutual Fund Analytics Platform for Bluestock Fintech. The platform ingests 10 public Indian mutual fund datasets (AMFI, mfapi.in, NSE/BSE), processes them through a Python ETL pipeline, loads 107,779 rows into a normalised SQLite star schema, and exposes insights through Power BI dashboards and Python analytics notebooks.

₹62.7L Cr

Total Industry AUM

₹31,002 Cr

SIP ATH Dec 2025

26.12 Cr

Total Folios

107,779

DB Rows Loaded

40

Funds Analysed

Key Findings at a Glance

- SBI Mutual Fund leads with ₹12.5L Cr AUM (19.9% market share)
- SIP inflows grew 169% from ₹11,517 Cr (Jan 2022) to ₹31,002 Cr (Dec 2025)
- Industry folios doubled from 13.26 Cr to 26.12 Cr in 4 years
- 97.8% of SIP investors have gaps exceeding 35 days — high discontinuity risk
- Small Cap funds show 100x higher daily VaR than Liquid funds (−2.39% vs −0.02%)
- ICICI Pru Midcap scored highest on composite scorecard (85.1/100)

2. Project Overview & Architecture

The platform follows a classic data engineering architecture: raw data ingestion, transformation, storage in a relational database, analytics, and visualisation. The 7-day development roadmap structured each deliverable as an independent, verifiable milestone.

7-Day Development Roadmap

Day	Focus	Key Deliverable
1	Data Ingestion	10 CSVs loaded, AMFI code validation, live NAV fetch
2	ETL & Database	SQLite star schema, 107K rows, 10 SQL queries
3	EDA	13 charts: NAV trends, demographics, sector allocation
4	Performance Metrics	CAGR, Sharpe, Sortino, Alpha, Beta, Scorecard
5	BI Dashboard	Power BI 4-page dashboard
6	Advanced Analytics	VaR/CVaR, cohort analysis, SIP continuity, HHI, Recommender
7	Final Report	This report + 12-slide PowerPoint presentation

Technology Stack

Layer	Tools & Libraries
Language	Python 3.14 — Pandas, NumPy, SciPy, SQLAlchemy, Matplotlib, Seaborn, Plotly
Database	SQLite3 — star schema with 2 dimensions and 9 fact/reference tables
BI Tools	Power BI Desktop (.pbix) — 4-page interactive dashboard
Analytics	scipy.stats.linregress (OLS regression), numpy percentile (VaR/CVaR)
Version Control	Git/GitHub — github.com/Shahinshafi1717/mutual-fund-platform

3. Data Pipeline & ETL

The ETL pipeline (etl_pipeline.py) reads all 10 raw CSVs, applies targeted cleaning rules per dataset, and saves cleaned outputs to data/processed/. The database loader (db_load.py) then creates the SQLite star schema and populates all 11 tables.

Database Schema — Table Inventory

Table	Type	Rows	Description
dim_fund	Dimension	40	40 mutual fund schemes with AMC, category, risk
dim_date	Dimension	1,826	Full calendar spine 2022-2026
fact_nav	Fact	64,320	Daily NAV after forward-fill for holidays
fact_transactions	Fact	32,778	Investor buy/sell/SIP transactions
fact_performance	Fact	40	Pre-computed and validated risk metrics
fact_portfolio	Fact	322	Equity holdings with sector weights
fact_aum	Fact	90	Quarterly AUM by fund house
fact_sip_industry	Fact	48	Monthly SIP inflows and active accounts
fact_category_inflows	Fact	144	Net inflows by fund category
fact_folio_count	Fact	21	Quarterly industry folio counts
benchmark_indices	Reference	8,050	7 benchmark indices daily closing values

Key Cleaning Operations

- NAV forward-fill: 18,320 weekend/holiday gaps filled using last known NAV value
- AMFI code integrity: cross-validated all 40 codes across 4 fact tables
- SIP YoY nulls: first 12 months imputed using 12-period lagged pct_change()
- Benchmark normalisation: 'NIFTY 100 TRI' mapped to 'NIFTY100' for joins
- Transaction validation: amount > 0, valid KYC enums, standardised tx_type casing

4. Exploratory Data Analysis

13 charts were generated across 5 themes: NAV trends, AUM growth, SIP inflows, investor demographics, and sector allocation. All charts were produced using Matplotlib, Seaborn, and Plotly from raw data stored in the SQLite database.

EDA Themes & Charts Produced

Theme	Charts	Key Visual
NAV Trends	3	Line chart — top 5 funds indexed to 100
AUM Growth	2	Clustered bar — fund house AUM comparison
SIP Inflows	3	Area chart with ATH annotation (Dec 2025)
Investor Demographics	3	Age/gender pie + SIP amount box plot
Sector Allocation	2	Heatmap — category net inflows FY2024-25

EDA Insights

- SBI Mutual Fund leads with ₹12.5L Cr AUM — 19.9% market share in latest quarter
- SIP inflows show strong upward trend: ₹11,517 Cr (Jan 2022) to ₹31,002 Cr (Dec 2025)
- Liquid category receives ₹34,000–42,000 Cr monthly — dominated by institutional investors
- 26–35 male cohort is the largest investor segment by total transaction value (₹64 Cr)
- Punjab leads state-wise investment at ₹315.8 Cr, followed by Tamil Nadu at ₹315.2 Cr
- Industry folios doubled from 13.26 Cr to 26.12 Cr — equity drives 75%+ of growth
- Banking sector holds 20% of all equity portfolio weight across 30 funds (₹62,840 Cr)
- Large Cap funds show pairwise correlation of 0.85–0.97 — minimal diversification benefit

5. Performance Metrics & Risk Analysis

All risk and return metrics were computed from raw NAV data using standard financial formulae. No pre-computed values were used from the source files.

Formulae Used

Metric	Formula
Daily Return	$r_t = (NAV_t / NAV_{t-1}) - 1$
CAGR	$(NAV_{end} / NAV_{start})^{(1/n)} - 1$ [n = years]
Sharpe Ratio	$(R_p - R_f) / Std(R_p) \times \sqrt{252}$ [Rf = 6.5% RBI repo]
Sortino Ratio	$(R_p - R_f) / Std_{downside}(R_p) \times \sqrt{252}$
Alpha & Beta	OLS regression via <code>scipy.stats.linregress</code> vs benchmark
Max Drawdown	$\min(NAV_t / \text{rolling_max}(NAV) - 1)$
VaR(95%)	5th percentile of daily return distribution
CVaR(95%)	Mean of returns below VaR threshold
HHI	$\sum(\text{weight}_i / 100)^2$ per fund portfolio

Top 10 Funds — Composite Scorecard

#	Fund	Score	3yr CAGR	Sharpe	Alpha	Expense
1	ICICI Pru Midcap Fund	85.1	31.8%	0.88	29.26%	1.36%
2	Axis Midcap Fund	82.0	35.1%	0.73	26.08%	1.38%
3	HDFC Mid-Cap Opportunities	80.5	32.4%	0.81	27.20%	1.38%
4	Mirae Asset Large Cap Fund	80.0	34.0%	1.07	26.98%	1.46%
5	Kotak Flexicap Fund	78.8	29.6%	0.97	27.79%	1.45%
6	SBI Small Cap Fund	75.8	26.7%	0.71	31.51%	1.43%
7	DSP Small Cap Fund	75.6	27.0%	0.71	29.35%	1.52%
8	ICICI Pru Bluechip Fund	75.1	32.5%	0.71	21.19%	0.80%
9	Mirae Asset Tax Saver Fund	73.4	29.2%	0.92	27.51%	1.60%
10	SBI Bluechip Fund	72.4	30.5%	0.86	23.20%	1.54%

Scorecard weights: 30% 3yr CAGR + 25% Sharpe Ratio + 20% Alpha + 15% Expense Ratio (inverse) + 10% Max Drawdown (inverse)

6. Advanced Analytics

6.1 Value at Risk (VaR) & CVaR

Historical VaR(95%) computed as the 5th percentile of the daily return distribution. CVaR is the mean of all returns below the VaR threshold. Small Cap funds carry 100x higher daily risk than Liquid funds.

Fund	VaR 95% (daily)	CVaR 95% (daily)	Category
ABSL Small Cap Fund	-2.392%	-3.029%	Small Cap
Axis Small Cap Fund	-2.328%	-2.969%	Small Cap
SBI Small Cap Fund	-2.316%	-3.016%	Small Cap
Nippon India Small Cap Fund	-2.281%	-2.994%	Small Cap
DSP Small Cap Fund	-2.152%	-2.857%	Small Cap
ICICI Pru Liquid Fund	-0.024%	-0.031%	Liquid

6.2 SIP Continuity Analysis

Among 1,362 investors with 6+ SIP transactions, 97.8% (1,332) show an average gap exceeding 35 days between SIPs. The median gap is 64.7 days — more than double the expected monthly cycle. This represents a significant retention risk requiring proactive intervention.

Metric	Value	Implication
Eligible investors (6+ SIPs)	1,362	Analysis base
At-risk investors (gap > 35 days)	1,332 (97.8%)	Critical retention risk
Median gap between SIPs	64.7 days	2x expected monthly cycle
Consistent investors	30 (2.2%)	Only 1 in 45 investors are consistent

6.3 Fund Recommender System

Rule-based recommender using weighted scoring across 4 metrics: Sharpe (35%), 3yr Return (30%), low Expense Ratio (20%), Morningstar Rating (15%).

Risk Appetite	Top Recommendation	Sharpe	3yr Return	Expense
Low	ICICI Pru Liquid Fund - Regular	7.68	7.68%	0.74%
Moderate	HDFC Top 100 Fund - Regular	1.06	14.84%	1.55%
High	Kotak Emerging Equity - Regular	0.96	18.23%	1.56%

7. Dashboard & Visualisation

An interactive Power BI dashboard (.pbix) was built as the primary visualisation deliverable. The dashboard contains 4 pages with a total of 10+ interactive slicers, enabling dynamic filtering by fund house, category, plan type, risk rating, state, age group, and time period.

Power BI Dashboard — Page Inventory

Page	Charts	Slicers	Key Visuals
Industry Overview	2	2 (year, fund house)	AUM bar, SIP line, folio growth, category pie
Fund Performance	2	3 (category, plan, risk)	Risk-return scatter, sortable fund table
Investor Analytics	3	3 (state, age, city tier)	State bars, age/gender split, donut chart
SIP & Market Trends	3	2 (month, category)	SIP trend, category bar, category donut

Dashboard Access

- Power BI File: dashboard/MutualFund_Dashboard.pbix — open with Power BI Desktop (free)
- Google Drive: https://drive.google.com/file/d/1nncvt_HEofhLfjXS3cl440Z5glB-ytvH/view
- Demo Video: <https://drive.google.com/file/d/1L-2ZQ6yPJg5LytejppHoRR0Qn0o3MgmX/view>

8. Conclusions & Recommendations

The Mutual Fund Analytics Platform successfully demonstrates end-to-end data engineering and analytics capabilities across the complete lifecycle from raw data ingestion to business intelligence delivery.

Business Recommendations

Target 26-35 Age Group	This cohort drives 18% of total investment volume (₹64 Cr). Personalised Moderate-risk Large Cap Direct plan campaigns can significantly boost AUM growth.
Address SIP Discontinuity	97.8% at-risk rate demands automated SMS/email alerts when SIP gaps exceed 35 days. Early intervention reduces churn and improves long-term retention.
Promote Direct Plans	Direct plans deliver 0.6–1.0% lower expense ratios, compounding to 15–25% more wealth over 10 years. Platform should default to Direct plan recommendations.
Diversify Large Cap Allocations	High pairwise correlation (0.85–0.97) among Large Cap funds means minimal diversification benefit. Guide investors toward Flexi Cap or Multi Cap alternatives.
Monitor Sector Concentration	Axis Bluechip (HHI 0.206) and ABSL Small Cap (0.201) show dangerously high sector concentration. Flag these in risk advisories and suggest rebalancing.

Mutual Fund Analytics Platform — Bluestock Fintech Capstone 2026
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